User Interface & User Experience



PeTIK Jombang

Pesantren PeTIK II YBM PLN
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Yayasan Baitul Maal



Pengenalan Software Development Life Cycle (SDLC) & Tahapan Perancangan UIUX

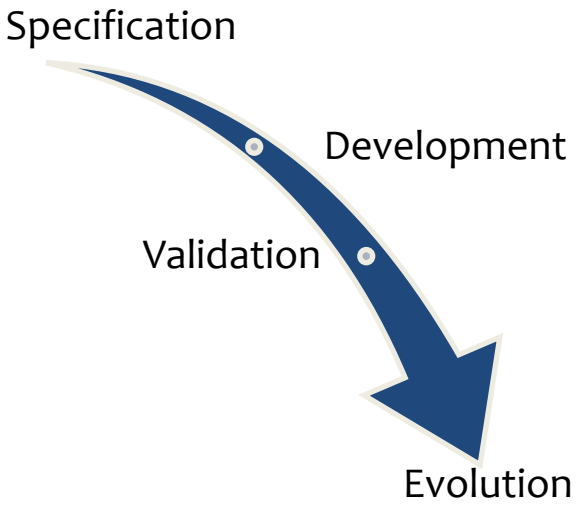
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2. Pengenalan Software Development Life Cycle (SDLC)

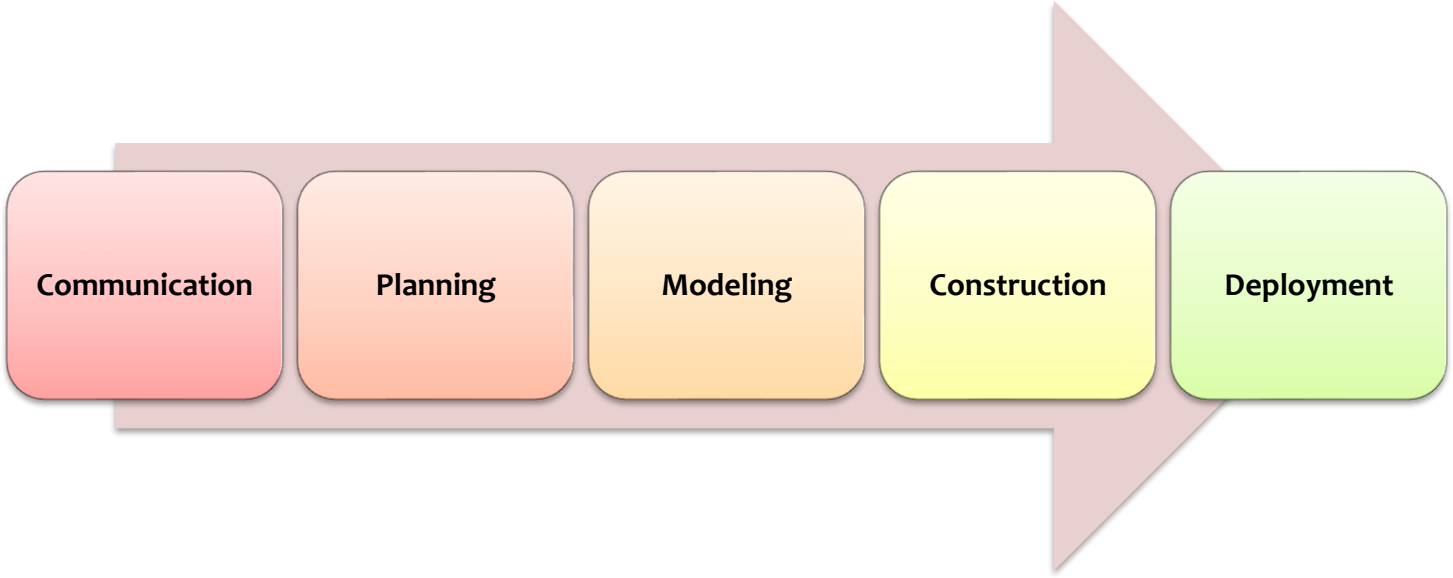
UI/UX

Software Engineering Process



- **Software specification,** where customers and engineers define the software that is to be produced and the constraints on its operation
- **Software development,** where the software is designed and programmed.
- **Software validation,** where the software is checked to ensure that it is what the customer requires.
- **Software evolution,** where the software is modified to reflect changing customer and market requirements

Software Process Framework



Software Process Framework

- **Communication**. Before any technical work can commence, it is critically important to communicate and collaborate with the customer (and other stakeholders).
- **Planning**. Any complicated journey can be simplified if a map exists. The map—***called a software project plan***—defines the software engineering work by describing the technical tasks to be conducted, the risks that are likely, the resources that will be required, the work products to be produced, and a work schedule.
- **Modeling**. Creating models to better understand software requirements and the design that will achieve those requirements.
- **Construction**. This activity combines code generation (either manual or automated) and the testing that is required to uncover errors in the code.
- **Deployment.** The software (as a complete entity or as a partially completed increment) is delivered to the customer who evaluates the delivered product and provides feedback based on the evaluation

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Prescriptive Model

UI/UX

Prescriptive Model

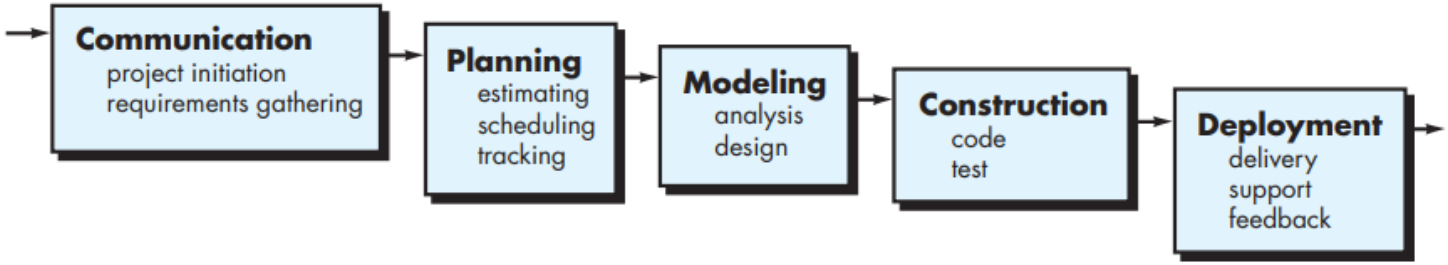
Definition:

Prescriptive process models define a prescribed set of process elements and a predictable process work flow.

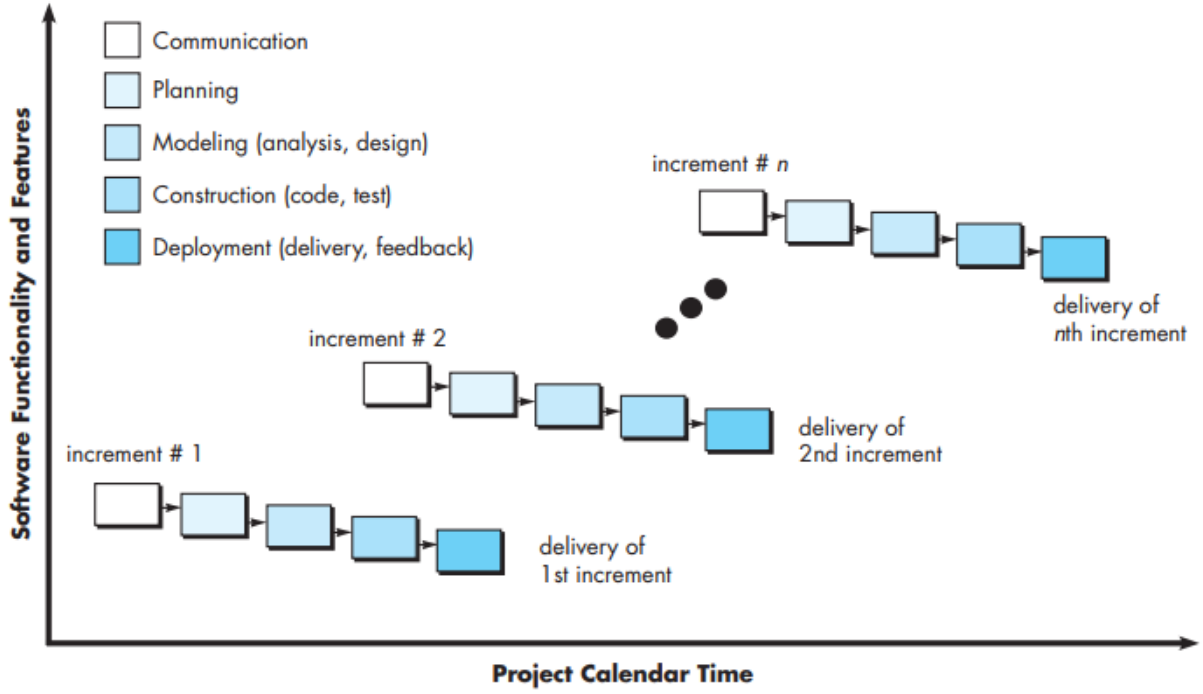
Example:

1. Waterfall
2. Incremental
3. Evolutionary: Prototyping
4. Concurrent Models

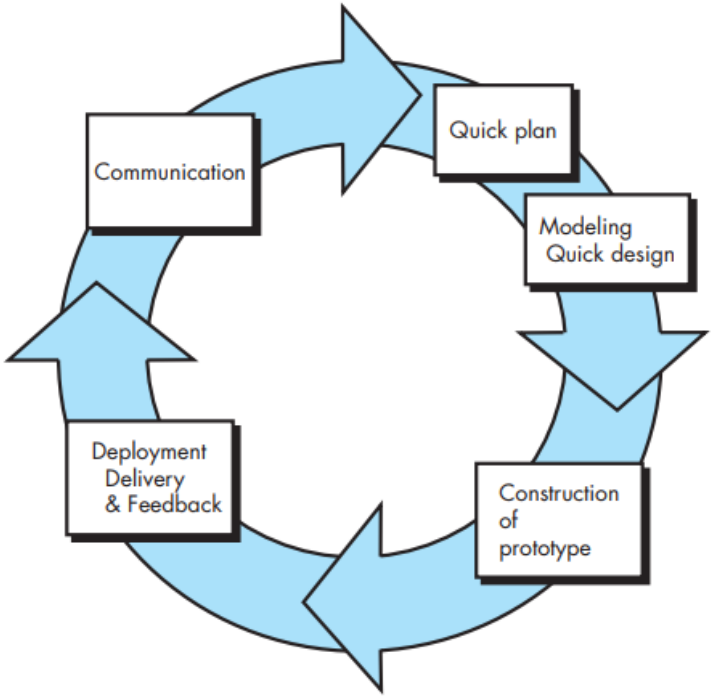
Prescriptive Model: Waterfall



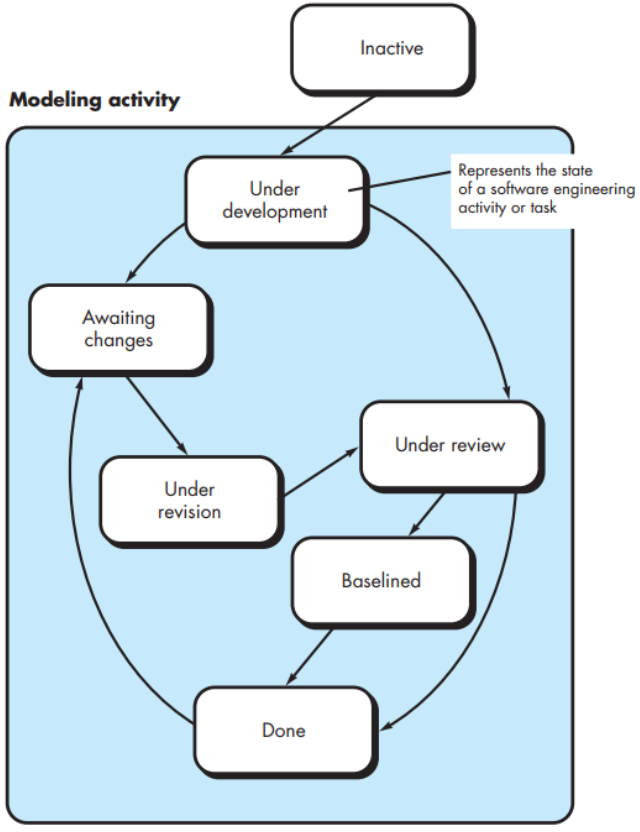
Prescriptive Model: Incremental



Prescriptive Model: Evolutionary, Prototyping



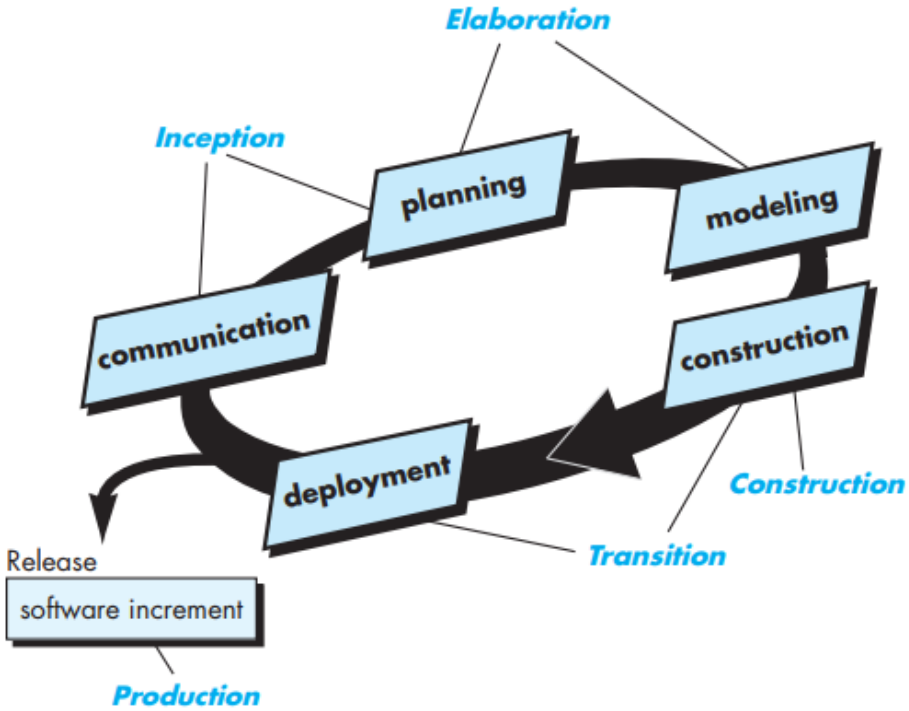
Prescriptive Model: Concurrent



Unified Process Model

- The ***inception*** phase of the UP encompasses both customer communication and planning activities. Fundamental business requirements are described through a set of preliminary use cases.
- The ***elaboration*** *phase* encompasses the communication and modeling activities of the generic process model
- The ***construction*** *phase* of the UP is identical to the construction activity defined for the generic software process.
- The ***transition*** *phase* of the UP encompasses the latter stages of the generic construction activity and the first part of the generic deployment (delivery and feedback) activity.
- The ***production*** *phase* of the UP coincides with the deployment activity of the generic process.

Unified Process



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Agile Model

UI/UX

Agile Manifesto

https://www.vanharen.net/blog/is-agile-new-agile-manifesto-allocated-out-of-the-agile-scrum-handbook/



Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and interactions	Over	processes and tools
Working software	Over	comprehensive documentation
Customer collaboration	Over	contract negotiation
Responding to change	Over	following a plan

That is, while there is value in the items on the right, we value the items on the left more.

Agile Manifesto

Nilai 1: *Individuals and interactions over processes and tools*

- Mengabaikan pentingnya individu dan interaksi adalah cara yang sangat cepat untuk gagal. Bagaimanapun, itu adalah orang-orang yang melakukan proyek. Beberapa manajer berpikir mereka dapat mengatasi masalah di bidang ini dengan menggunakan "sistem" yang lebih canggih, tetapi itu jarang berhasil.

Nilai 2: *Working software over comprehensive documentation*

- Ini merujuk pada fakta bahwa, alih-alih menggunakan dokumentasi dimuka untuk memprediksi apa yang perlu terjadi dalam suatu proyek, kita hanya melanjutkan, membuat bagian-bagian dari perangkat lunak yang berfungsi (kenaikan), dan menggunakannya untuk beradaptasi.

Agile Manifesto

Nilai 3: *Customer collaboration over contract negotiation*

- Setiap proyek akan lebih berhasil dengan tingkat kolaborasi pelanggan yang lebih tinggi. Pelanggan harus berkolaborasi dengan Anda setiap saat ketika Anda terus-menerus menentukan *requirement* baru dan meminta mereka untuk memeriksa *update* dan memberi Anda *feedback*.

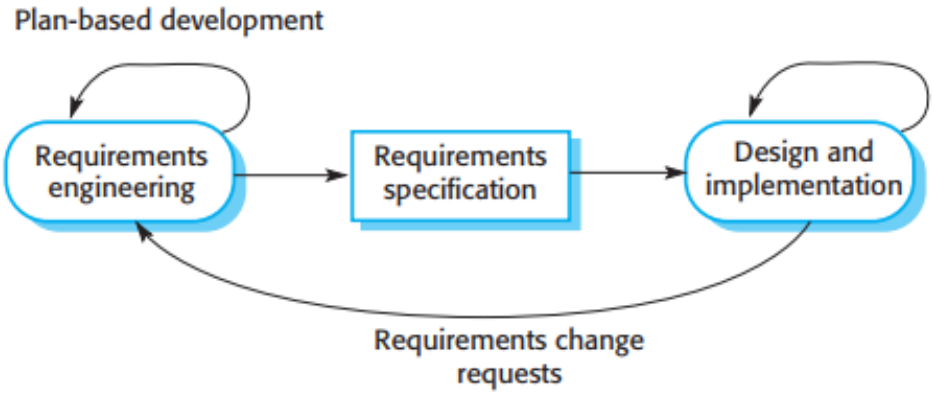
Nilai 4: *Responding to change over following a plan*

- Yang terakhir ini biasanya disebut sebagai "perubahan" di Agile, mungkin karena itu membuat pelanggan senang mengetahui bahwa mereka bebas untuk mengubah segalanya, tetapi pada kenyataannya, itu bukan perubahan kecuali jika itu tidak sesuai dengan rencana dasar awal.

Agile Methods - Characteristics

- Proses spesifikasi, desain, dan implementasi **saling terkait**. Tidak ada spesifikasi sistem yang terperinci, dan **dokumentasi** desain **diminimalkan** atau dihasilkan secara otomatis oleh lingkungan pemrograman yang digunakan untuk mengimplementasikan sistem
- Sistem ini dikembangkan dalam serangkaian *increment*. Pengguna akhir dan pemangku kepentingan sistem lainnya dilibatkan dalam menentukan dan mengevaluasi setiap *increment.*
- Dukungan alat yang luas digunakan untuk mendukung proses pengembangan. Alat yang dapat digunakan **termasuk alat pengujian otomatis**, alat untuk mendukung **manajemen konfigurasi**, dan **integrasi sistem** dan alat untuk **mengotomatisasi produksi antarmuka pengguna.**

Plan-driven vs Agile Development

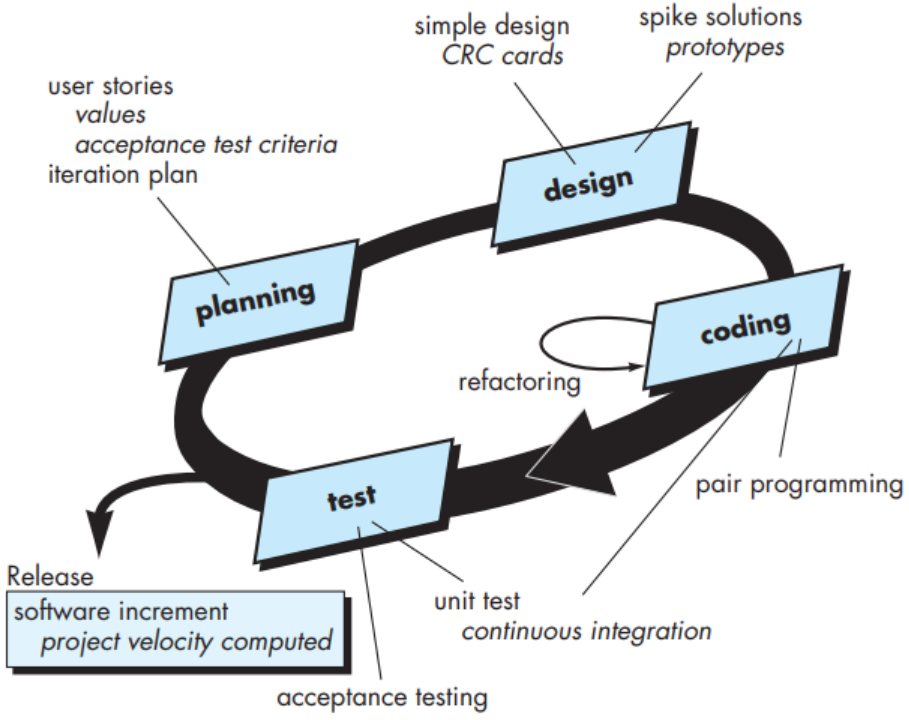


Metode Agile akan berhasil pada tipe proyek berikut:

- Product development where a software company is developing a **small or medium-sized product for sale**.
- Custom system development within an organization, where there is **a clear commitment from the customer** to become involved in the development process and where there are few external stakeholders and regulations that affect the software.

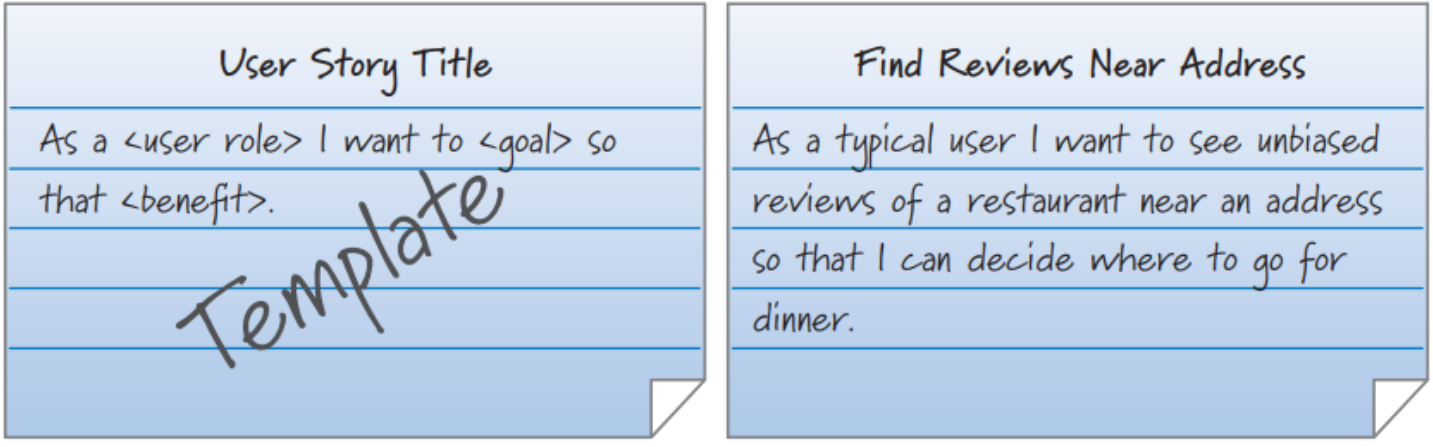


Extreme Programming (XP)



Extreme Programming (XP)

- **User Stories:** User stories are a convenient format for expressing the desired business value for many types of features.



Extreme Programming (XP)

http://agilemodeling.com/artifacts/crcModel.htm

- **CRC Cards:** is a collection of standard index cards that have been divided into **three** sections. A **class** represents a collection of similar objects, a **responsibility** is something that a class knows or does, and a **collaborator** is another class that a class interacts with to fulfill its responsibilities

Class Name	
Responsibilities	Collaborators

Student	
Student number Name Address Phone number Enroll in a seminar Drop a seminar Request transcripts	Seminar

Extreme Programming (XP)

- **Pair Programming:** The programming pair sits at the same computer to develop the software
  - ◇ It supports the idea of **collective ownership** and responsibility for the system (egoless programming)
  - ◇ It acts as an **informal review process** because each line of code is looked at by at least two people.
  - ◇ It **encourages refactoring** to improve the software structure



Scrum - Framework

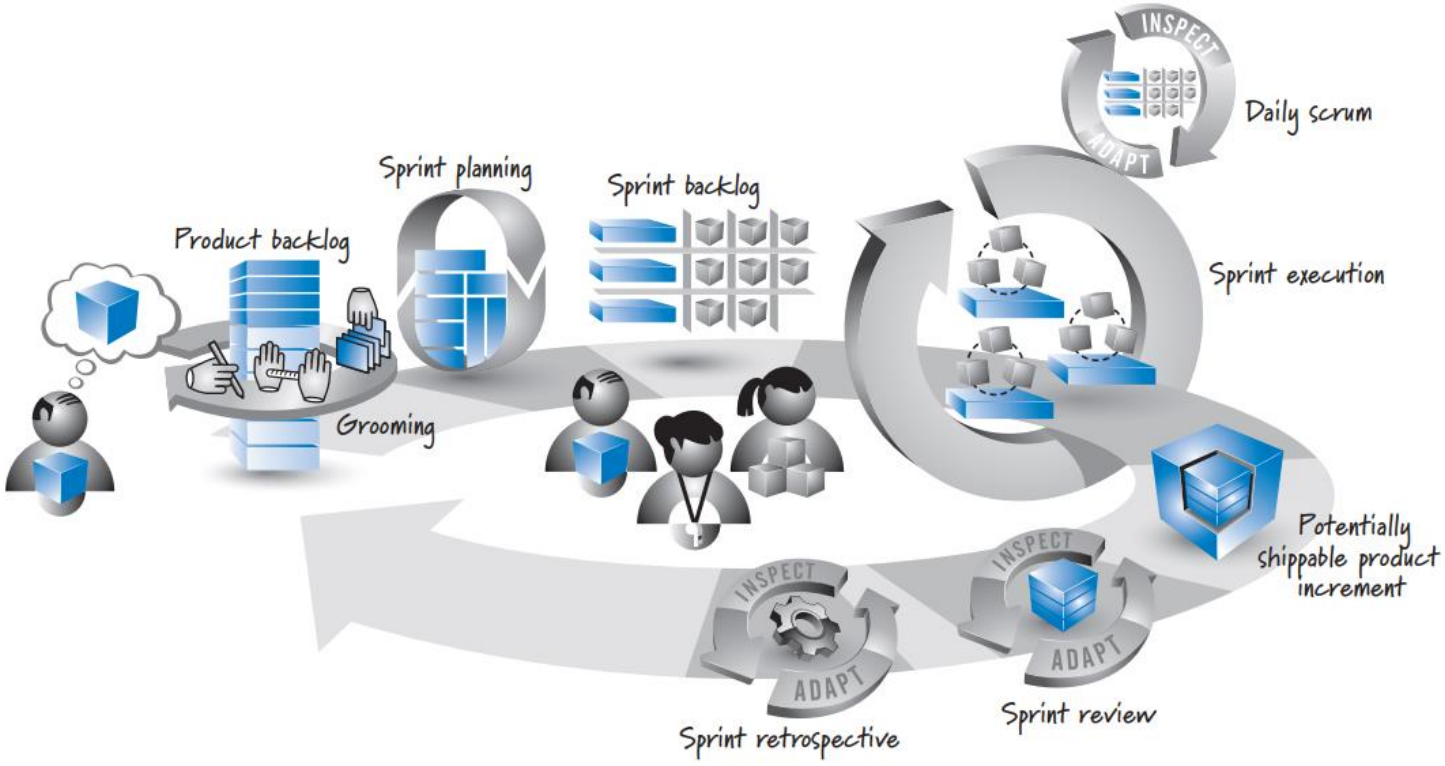
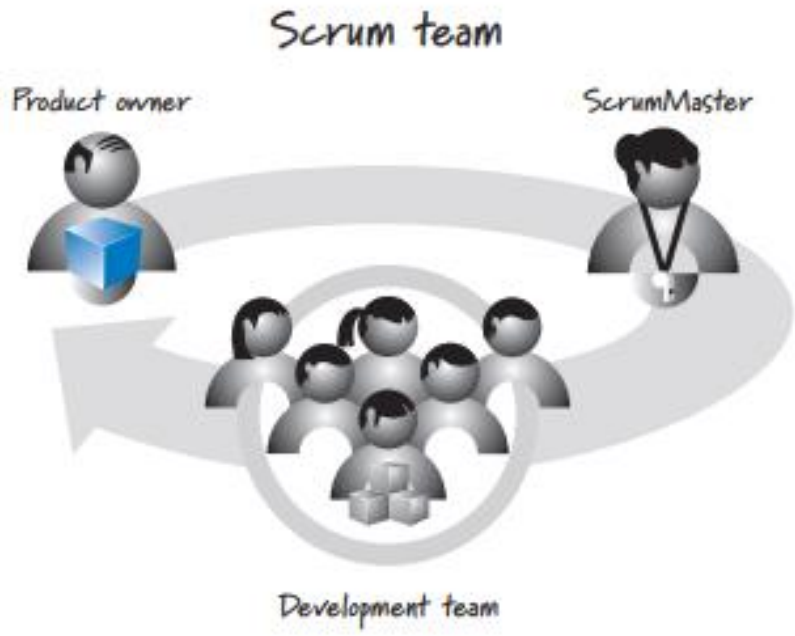


FIGURE 2.3 Scrum framework

Scrum - Roles

- The **product owner** is responsible for what will be developed and in what order
- The **ScrumMaster** is responsible for guiding the team in creating and following its own process based on the broader *Scrum framework*
- The **development team** is responsible for determining how to deliver what the product owner has asked for



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Tahapan Perancangan UI/UX

UI/UX

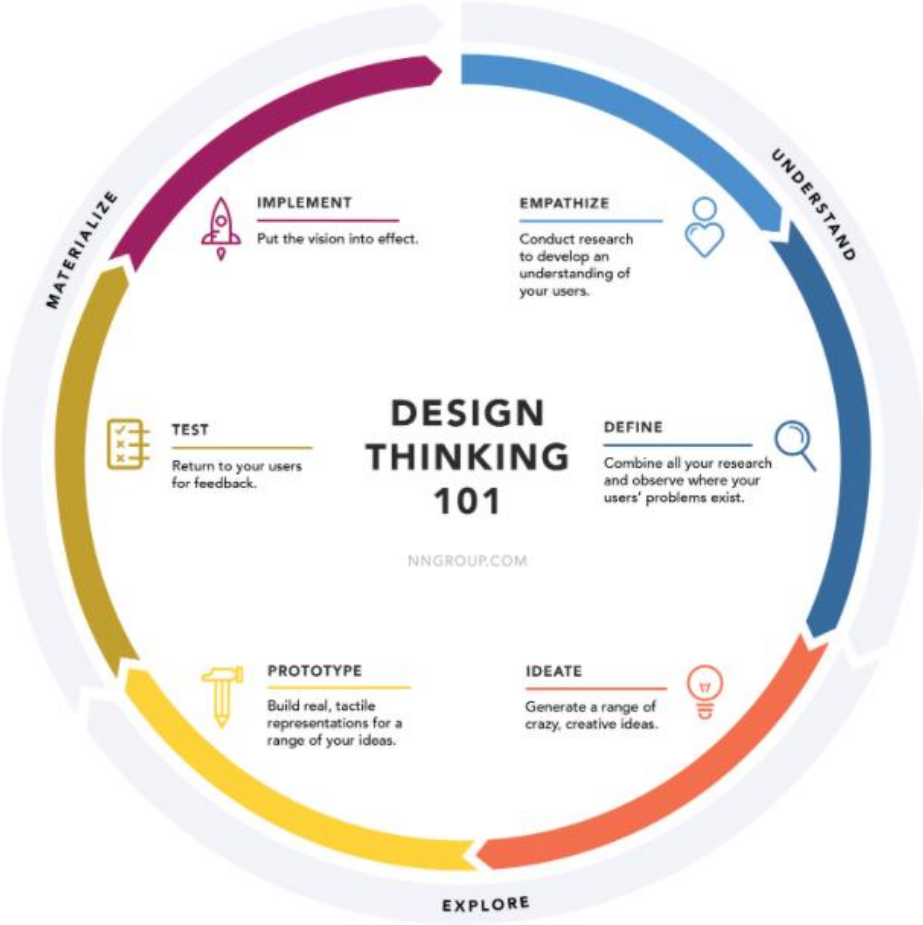
What is Design Thinking?



- Design thinking **is not an exclusive property of designers**—all great innovators in literature, art, music, science, engineering, and business have practiced it.
- Design thinking is an **iterative process** in which we seek to understand the user, challenge assumptions, and redefine problems in an attempt to identify alternative strategies and solutions that might not be instantly apparent with our initial level of understanding.

What is Design Thinking?

- The design thinking ideology asserts that a hands-on, user centric approach to **problem solving** can lead to **innovation**, and innovation can lead to **differentiation** and a **competitive advantage**.
- This hands-on, user-centric approach is defined by the design thinking process and comprises **6 distinct phases**, as defined and illustrated beside.



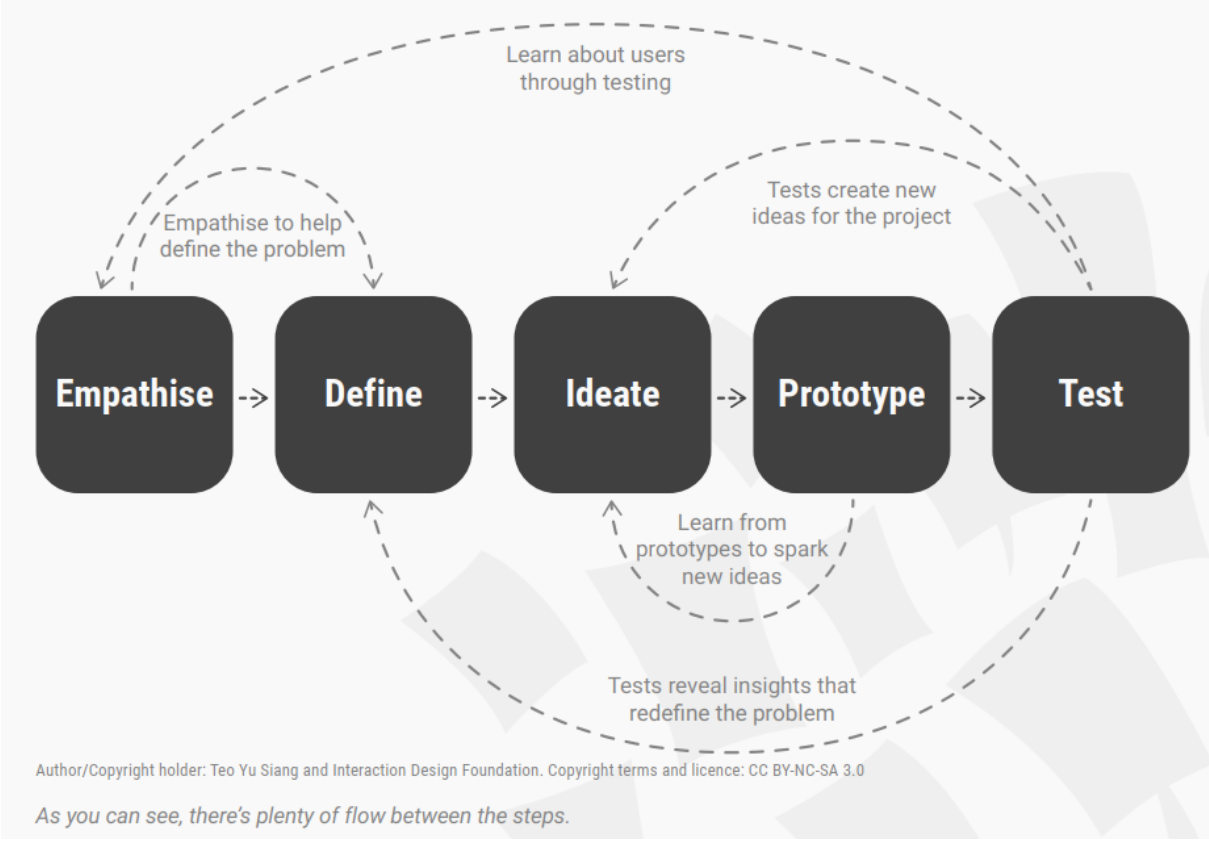
Design Thinking Phases

- **Empathize:** Conduct research in order to develop knowledge about what your users do, say, think, and feel.
- **Define:** Combine all your research and observe where your users' problems exist. In pinpointing your users' needs, begin to highlight opportunities for innovation.
- **Ideate**: Brainstorm a range of crazy, creative ideas that address the unmet user needs identified in the define phase. Give yourself and your team total freedom; no idea is too farfetched and quantity supersedes quality.

Design Thinking Phases

- **Prototype**: Build real, tactile representations for a subset of your ideas. The goal of this phase is to understand what components of your ideas work, and which do not. In this phase you begin to weigh the impact vs. feasibility of your ideas through feedback on your prototypes.
- **Test**: Return to your users for feedback. Ask yourself ‘Does this solution meet users’ needs?’ and ‘Has it improved how they feel, think, or do their tasks?’
- **Implement**: Put the vision into effect. Ensure that your solution is materialized and touches the lives of your end users.

Design Thinking is an Iterative and Non-linear Process



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As you can see, there's plenty of flow between the steps.

Design Thinking Process

- Design thinking is an iterative and non-linear process.
- This simply means that the design team continuously use their results to review, question, and improve their initial assumptions, understandings and results.
- Results from the final stage of the initial work process inform our **understanding of the problem,** help us **determine the parameters of the problem, enable us to redefine** the problem, and, perhaps most importantly, **provide** us with **new insights** so **we can see any alternative solutions** that might not have been available with our previous level of understanding

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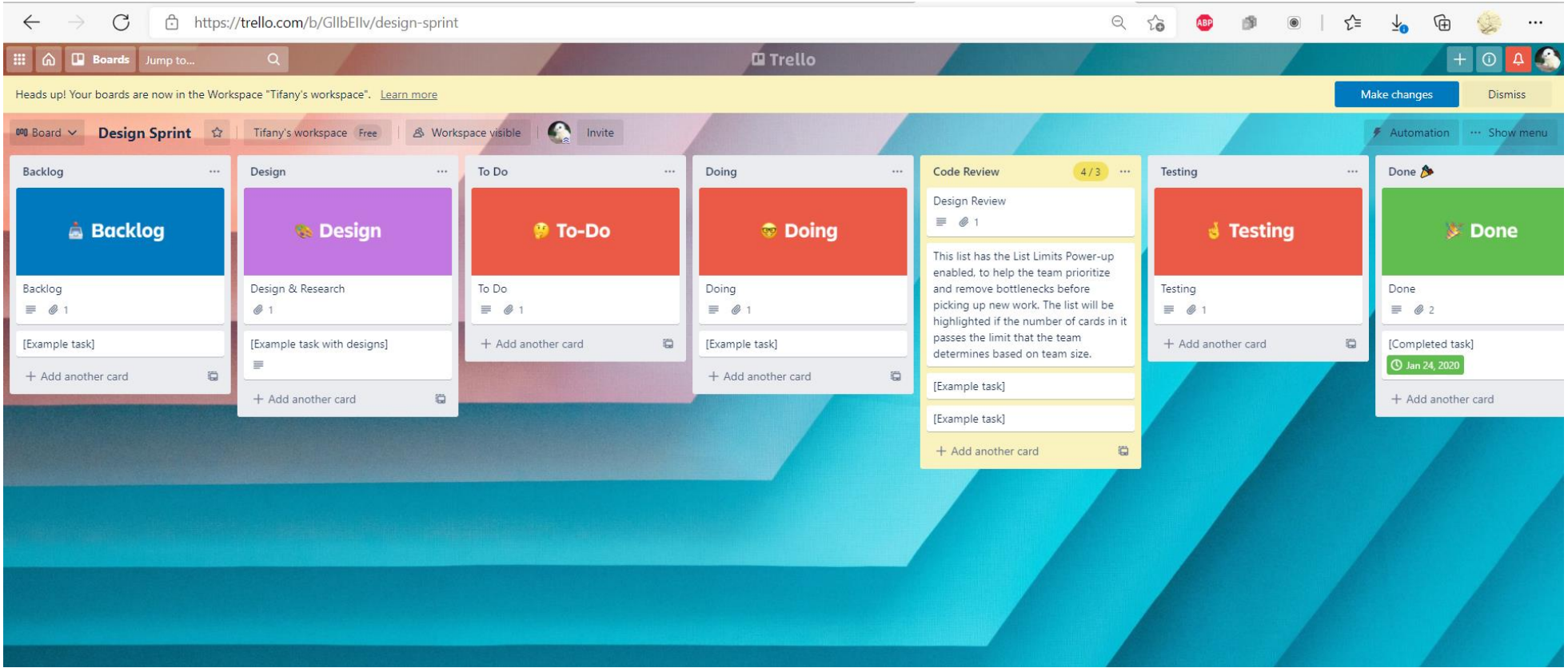
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Praktik Sprint Menggunakan Trello

UI/UX

Sprint pada Trello

https://trello.com/



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TERIMA KASIH
ATAS SEGALA PERHATIAN
SEMOGA BERMANFAAT...

UI/UX